

YOUR PC



Compare the performance of your system's memory subsystem with that of other machines

that need access to it. So as you can see, keeping an eye on how fast your memory works is absolutely critical.

To benchmark memory, startup SiSoft Sandra and you'll be presented with a number of icons, each designed to carry out a different test or task. Again, ensure that no other applications are running while the benchmark does its thing.

Double-click the icon which reads 'Memory Performance' and the program will begin the testing process. The test scores show a comparative summary between your PC's memory subsystem versus that of other machines.

What to expect

500 MHz CPU: 500 MBps

1 GHz CPU: 950 MBps

1.5 GHz CPU: 1500 MBps

Warning

To ensure the most accurate results, avoid moving the mouse or pressing any keys on the keyboard while the test is running. Also, to ensure that you are getting maximum marks for your rig, make sure that your motherboard drivers are installed properly. If

you're unsure about this, one quick way to check for gaps is by navigating to **Control Panel > System**. Under the 'Device Manager' tab, look out for any exclamation symbols under the System Devices tree. If one or a few happen to appear, reboot your system into Safe Mode and remove them by highlighting the entry and clicking 'Remove'. Reboot the computer and you will be prompted to install the correct drivers as necessary.

System Prerequisites

486 or higher CPU, 12 MB of RAM, 10 MB of hard disk space

THE HARD DISK SUBSYSTEM

This is perhaps the most underrated component of a PC. After the CPU and memory, the hard disk is the next determining factor in how fast your system works. This is because a lot of information required to run applications is not loaded in the physical memory (RAM) and needs to be read from the disk. HDTach is an extremely powerful tool which rigorously investigates hard disk activity across a series of parameters from seek times to transfer rates. It is fast becoming the *de facto* standard in fixed storage performance testing.

Install HDTach and close all running programs before launching the test. Usually, hard disk benchmark tools need plenty of space for temporary files,

a shortage of which could yield highly inaccurate results. Before you start the examination, be sure that HDTach inspects a partition with at least 500 MB of free space. If your hard disk is partitioned into more than one segment, use the part that doesn't hold the Windows operating system.

Once you've selected the right partition for testing, begin the test. The test could take up to 45 minutes depending on your configuration.

Another way to check hard disk performance is through SiSoft Sandra, the Swiss-army knife of benchmarking tools. Double-click the 'Drive Performance' icon. Under Drive Selection, choose the partition you wish the program to test. The test should begin and last approximately five minutes.

What to expect

7200-rpm drive: Average of 35 MBps bandwidth, 11.8 ms or lower access time

5400-rpm drive: Average of 25 MBps bandwidth, 13.1 ms or lower access time

Warning

HDTach doesn't work too well with RAID0 arrays when the Advanced Size Check is



SiSoft Sandra 2001 TE, 3DMark 2001 SE, HDTach, Quake III: Arena, Evolve v938, Serious Sam SE and more